MD Shihabul Islam Shovo

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Professional Summary

Computer Science and Engineering graduate combining hands-on full-stack software development with active peer-reviewed research in applied deep learning. Has designed and shipped production web applications using Django REST Framework, React, ASP.NET MVC, and Flask, building REST APIs, JWT authentication, and role-based access control, while also co-authoring **11** peer-reviewed and in-progress papers spanning medical imaging (dermatology, ophthalmology, radiology), wearable biosignal analysis (EEG, IMU), parameter-efficient adaptation of foundation models, and the reliability of agentic vision-language systems. Comfortable operating across the full lifecycle of a piece of work: requirements, architecture, implementation, testing, and deployment on the engineering side; problem definition, experimentation, statistical validation, and technical writing on the research side. Currently deepening backend engineering skills in C#, ASP.NET, and SQL through an intensive Servercamp program.

Research Interests

Medical Image Analysis · Lightweight & Efficient Deep Learning · Semi-Supervised Learning · State-Space Models (Mamba) · Biosignal & Wearable Sensing · Explainable AI · Parameter-Efficient Adaptation · Trustworthy Vision-Language Models

Education

B.Sc. in Computer Science and Engineering — *2020 – 2024*
Ahsanullah University of Science and Technology (AUST) — *Dhaka, Bangladesh*
CGPA: **3.078 / 4.00**

Higher Secondary Certificate (HSC) — *2016 – 2018*
Cantonment Public School & College, Saidpur — *Rangpur, Bangladesh*
GPA: **5.00 / 5.00**

Secondary School Certificate (SSC) — *2014 – 2015*
Cantonment Public School & College, Saidpur — *Rangpur, Bangladesh*
GPA: **5.00 / 5.00**

Technical Skills

AI/ML & Research	PyTorch, TensorFlow, Keras, Scikit-learn, Transformers, Mamba / state-space models, Grad-CAM & explainable AI, multi-seed statistical validation (Wilcoxon, ANOVA), structural reparameterization
Foundation Models	LoRA / parameter-efficient fine-tuning, vision transformers (DINOv2, CLIP, SAM), vision-language models (Qwen2.5-VL, LLaVA-Med), BiomedCLIP retrieval, tool-using (ReAct) agents, linear CKA representation analysis
Languages	Python, JavaScript, TypeScript, C#, Java, C/C++
Frontend	React, Tailwind CSS, Bootstrap, HTML, CSS
Backend	Django, Django REST Framework, ASP.NET MVC, Flask, PHP
Databases	SQLite, MySQL, PostgreSQL, Microsoft SQL Server
Tools & Auth	Git, GitHub, Postman, Selenium WebDriver, JWT Authentication, Swagger/OpenAPI
Concepts	REST APIs, MVC Architecture, RBAC, Data Structures & Algorithms, SDLC, Agile Methodology

Professional Experience

Web & Automation Engineer, *Cleaning Connected* — *Jan 2025 – Present*
California, USA
(Remote) · Part-time

- Designed and developed digital solutions supporting business operations, including website development, lead management automation, and workflow optimisation.
- Maintained web platforms, integrated business tools, and resolved technical issues to improve operational efficiency.

Research Experience

ELITE Research Lab LLC — *Jun 2026 – Present*
Junior Research Assistant (Aug 2026 – Present); Research Intern (Jun 2026 – Aug 2026) — *Queens, NY, USA*
(Remote) · Part-time

Project: LCM-UNet – Lightweight Skin Lesion Segmentation using a Mamba-based U-Net

- Designed and led development of an ultra-lightweight (∼34K-parameter, 0.06 GFLOPs) Mamba-based state-space U-Net for melanoma-related lesion boundary detection, targeting deployment on low-cost point-of-care hardware.

- Proposed the Residual Dilated Local Adapter (RDLA), a zero-initialized dilated convolution module that adds optimisation capacity during training and is exactly fused back into the base kernel via structural reparameterization before inference, at zero added deployment cost.
- Validated the approach with multi-seed statistical testing (image-level Wilcoxon, $p < 10^{-4}$) on ISIC 2017, ISIC 2018, and external PH2 datasets, demonstrating consistent boundary-quality improvements over the baseline.
- Promoted from Research Intern to Junior Research Assistant in recognition of research contributions.

NIMISHES Lab, *Lead Researcher* *2026 – Present*

- Independent research lab building explainable, deployment-efficient AI for clinical use; contributed to 10 manuscripts across medical imaging (dermatology, ophthalmology, radiology), wearable biosignal analysis (EEG, IMU), and underwater computer vision, including first-author AquaProbe with the University of South Florida and APED-CXR with the American University of Sharjah.
- Extended the lab's work into parameter-efficient adaptation of frozen foundation models (LoRA rank allocation for vision transformers) and the reliability of agentic vision-language systems for medical imaging.
- Contributed to model design, cross-validation and statistical significance testing, and Grad-CAM-based explainability analysis across projects, consistently achieving state-of-the-art parameter efficiency versus standard backbones (e.g. ResNet50, DenseNet201).

Publications

***Accepted (4)**: peer-reviewed and accepted for presentation at IEEE conferences and IEEE/CVF workshops*

[C1] M. S. I. Shovo, N. K. Paul, T. Dong, and X. Lin, "AquaProbe: Efficient Adaptation of Frozen Vision Models for Underwater Images," *5th Workshop on Maritime Computer Vision (MaCVi), IEEE/CVF Winter Conf. on Applications of Computer Vision (WACV) 2027*, accepted. *(First author)* arXiv
Result: degradation-conditioned per-block CKA probe over 40 physically rendered Jerlov water-type/range conditions sets per-block LoRA ranks in a frozen ViT; 70.1 mIoU on SUIM, +1.9 mIoU over uniform allocation at the same 0.59M adapter budget (uniform needs 4× the parameters to match); +1.5 mIoU on SAM ViT-B and on TrashCan.

[C2] M. S. I. Shovo, N. K. Paul, A. Rahman, and I. J. Esha, "WaveFoG: Wavelet-Gated Transformer for Parkinson's Freezing of Gait Detection Using Wearable Accelerometer Signals," *5th IEEE Int. Conf. on Biomedical Engineering, Computer and Information Technology for Health (BECITHCON)*, Dhaka, Bangladesh, 4–5 Sept. 2026. *(First author)* arXiv · Code
Result: F1 0.875 ± 0.017, AUPRC 0.833 ± 0.020 on the Kaggle TLVMC FoG dataset; up to +5.4pp F1 over 1D CNN, Bi-LSTM, and vanilla Transformer baselines under subject-independent grouped 10-fold CV; temporal saliency maps align with reported FoG onset characteristics.

[C3] N. K. Paul, **M. S. I. Shovo**, I. J. Esha, and A. Rahman, "SleepEffFormer: Efficient CNN-Transformer with Transition-Aware Smoothing for Single-Channel EEG Sleep Stage Classification," *5th IEEE BECITHCON*, Dhaka, Bangladesh, 4–5 Sept. 2026. arXiv · Code
Result: 83.9% accuracy, 78.9% macro-F1, $\kappa = 0.765$ on Sleep-EDF Expanded (78 recordings); matches AttnSleep with 3–5× fewer parameters (∼367K); transformer encoder adds +6.6pp macro-F1, TAS adds +1.8pp with zero trainable parameters.

[C4] NIMISHES Lab, "DR-LiteNet: A Lightweight Explainable Hybrid CNN Framework for Imbalanced Diabetic Retinopathy Grading via Adaptive SMOTE Fusion," *1st Int. Conf. on Next-Generation Electrical & Electronics, Computer Systems, and Technologies (iCONEECT)*, 2026. arXiv
Result: $88.77 \pm 0.61\%$ accuracy, QWK 0.884 ± 0.010, macro-F1 0.792 ± 0.008 on APTOS 2019 with only 4.69M parameters (5× smaller than ResNet50, ΔQWK $= 0.022$); two-phase training with Grad-CAM explainability.

***Under Review (6)**: 1 to IEEE JBHI, 3 to IEEE ISBI 2027, 1 to IEEE/CVF WACV 2027 FABRIC, 1 to IEEE ICCIT 2026*

[U1] M. S. I. Shovo, N. K. Paul, and K. Morol, "LCM-UNet: A Reparameterizable Local-Compensated Mamba U-Net for Skin Lesion Segmentation," *IEEE Journal of Biomedical and Health Informatics (JBHI)*, under review. *(First author, IF: 7.7, Q1)* arXiv
Result: 34,180 parameters, 0.05974 GFLOPs after reparameterization (4 parameters more than the reference, identical cost); +0.48/+0.37pp IoU over a retrained baseline on ISIC 2017/2018; external PH2 transfer $86.37 \pm 0.29\%$ IoU vs. $85.91 \pm 0.29\%$; Wilcoxon significant at $p < 10^{-4}$.

[U2] M. S. I. Shovo, N. K. Paul, and J. Fayyad, "APED-CXR: Adaptive Privileged Evidence Distillation for Chest X-Ray Grounding with Image-Only Inference," *IEEE Int. Symp. on Biomedical Imaging (ISBI) 2027*, under review. *(First author)* arXiv
Result: an 11.1M-parameter image-only student distills report-informed spatial evidence from a frozen multimodal teacher verifying current student proposals; PadChest-GR micro-gF1 0.338 ± 0.018 vs. 0.302 ± 0.005 with fixed proposals (IoU ≥ 0.5; paired gain 0.036, 95% CI [0.022, 0.049]); macro-AUC 0.815; 5.7 GMACs and 9.6 ms/image on an A100 (batch 1). Teacher and reports are training-only.

[U3] N. K. Paul, **M. S. I. Shovo**, and D. Mahapatra, "Band2Net: Frequency-to-Network Transport Distillation for fMRI-Privileged, EEG-Only Session-Condition Decoding," *IEEE Int. Symp. on Biomedical Imaging (ISBI) 2027*, under review. arXiv

[U4] N. K. Paul, **M. S. I. Shovo**, and D. Sidibé, "LAPIS: Clinically Constrained Patient-Identity Residual Learning for Chest X-Ray Anonymization," *IEEE Int. Symp. on Biomedical Imaging (ISBI) 2027*, under review. arXiv

[U5] N. K. Paul, **M. S. I. Shovo**, M. M. Rahman, and J. Fayyad, "Coupled or Independent? Fault Coupling in Agentic Medical Vision Systems," *Workshop on Foundation AI for Biomedical Reasoning, Imaging, and Cognition (FABRIC), IEEE/CVF WACV 2027*, under review. arXiv
Result: coupling taxonomy with a provable detection limit for hard-label consistency checks; on PAD-UFES-20 and external HIBA dermatology data with two 7B VLM reasoners (Qwen2.5-VL, LLaVA-Med), tool order alone turns detectable disagreement into hidden joint wrong agreement (55% vs. 35% of cases; disagreement recall 0.29 vs. 0.57).

[**U6**] NIMISHES Lab, "AWEF-Net: An Attention-Weighted Ensemble Fusion Framework for Multi-Class Pulmonary Disease Classification from Chest Radiographs," *29th IEEE ICCIT*, 2026, under review. PDF
Result: 96.3% macro-F1, 97.1% COVID-19 recall on the COVID-19 Radiography Database (21,165 images, 4 classes); +1.4pp macro-F1 and +1.7pp COVID recall over the strongest fixed-fusion baseline; adaptive attention gate adds under 0.6% of total model parameters.

***In Preparation (1)**: planned submission to MIDL 2027*

[**P1**] N. K. Paul, **M. S. I. Shovo**, and C. Gonzalez (Medical University of Vienna), "CoMAF-Polyp: Calibrated Reliability-Aware Fusion of Specialist and Foundation Pseudo-Labels for Semi-Supervised Polyp Segmentation," planned submission to *Medical Imaging with Deep Learning (MIDL) 2027*.
Result: 88.46% Dice on Kvasir-SEG with 10% labels, recovering 98.6% of a per-image oracle; +2.44pp average Dice over the best baseline across four unseen datasets (ClinicDB, ColonDB, ETIS, CVC-300) under domain shift; calibration reduces ECE from 0.147 to 0.038; deployed model runs at 38.2 FPS.

Software Development Projects

Hospital Management System, *Full Stack (DRF, React, JWT, SQLite/PostgreSQL)* GitHub

- Designed and built a role-based hospital management platform (admin, doctor, patient, receptionist) with a Django REST Framework API and a React (Vite) SPA, modeling 9 relational entities including patients, doctors, appointments, prescriptions, and billing.
- Implemented JWT authentication with automatic silent token refresh (with request deduplication) and dual-layer role-based authorization that combines queryset-level scoping with object-level permission checks, ensuring users can only access their own records.
- Built a consistent API error-handling layer (custom DRF exception handler) with auto-generated OpenAPI/Swagger docs via drf-spectacular, reusable React data-fetching hooks, protected/role-gated routing, and a centralized axios client with request/response interceptors.

Multi-Tenant SaaS Project Management Platform, *Django REST Framework, PostgreSQL, Celery, Redis, Docker, JWT, React* GitHub

- Architected and developed a scalable multi-tenant project management API using Django REST Framework with organization-level data isolation, JWT authentication, and role-based permissions for administrators, managers, and team members.
- Implemented asynchronous background task processing with Celery and Redis for notifications, activity tracking, and scheduled operations, decoupling long-running tasks from the request cycle.
- Optimized database performance using Django ORM query optimization, strategic indexing, cursor-based pagination, and efficient relational modeling on PostgreSQL with support for millions of records.
- Designed API features including resource filtering, full-text search, rate limiting, and request throttling; documented all endpoints with auto-generated Swagger/OpenAPI specifications; containerized the full stack with Docker (PostgreSQL, Redis, Celery) for consistent local development and production deployment.

EasyHome: House Rentals Service, *ASP.NET MVC, C#, Entity Framework, MSSQL, Bootstrap* GitHub

- Built a complete house rental management system for landlords and tenants with role-based access, apartment dashboards, and rent requests.
- Implemented reviews, ratings, and an in-app chat system alongside user registration and profile management.

MRI Tumor Detection System, *Python, Flask, TensorFlow, Keras, VGG16* GitHub

- Developed a web-based AI system to classify MRI brain scans into four tumor categories.
- Implemented confidence scoring and automated PDF medical report generation, and optimized a CNN-based model for medical image classification.

Interactive Teaching Platform, *Django, DRF, Tailwind CSS, JavaScript, SQLite* GitHub

- Built an interactive learning platform with clickable terms, multimedia popups, and a responsive UI, backed by REST APIs for dynamic content loading.
- Implemented multilingual slug logic with Bengali fallback for SEO-friendly URLs.

NOTES App, *Django REST Framework, React, SQLite, JWT* GitHub

- Implemented user-specific CRUD operations with protected routes, integrating frontend and backend via REST APIs and CORS configuration.

Exclusive Shop, *React, TypeScript, Tailwind CSS* GitHub

- Implemented authentication, cart, wishlist, and responsive design, following component-based architecture and clean state handling.

Lift: E-commerce Platform, *PHP, JavaScript, HTML, CSS, MySQL, AJAX* GitHub

- Developed a full-featured grocery e-commerce website with user and admin panels, including product listing, cart, checkout, user authentication, and a reporting system.
- Built an admin dashboard to manage products, users, and customer feedback.

Thesis & Independent Research (Proposals)

ShinovaNet: Multi-View Hybrid CNN-Transformer with Cross-Attention Fusion and Explainable AI for Breast Cancer Detection, *Proposed Research*

- Proposed a dual-branch, multi-view architecture combining ResNet50 and Swin-B Transformer to extract local lesion morphology and global mammographic context from paired CC/MLO views.
- Designed learnable cross-attention fusion for intra-view and inter-view semantic alignment, integrating Grad-CAM and Attention Rollout for explainability and BI-RADS-aligned report generation, with cross-dataset evaluation on CBIS-DDSM and VinDr-Mammo.

Study on Benchmark Datasets for Human Disease Genes, *Academic Research*

- Analyzed gene–disease associations using benchmark biomedical datasets including DisGeNet, OMIM, CTD, and GWAS.
- Evaluated machine learning models including Decision Trees, Random Forests, SVM, and KNN for classification performance, and explored genomic data extraction tools including BioBERT and PubTator.

Certifications and Professional Development

Artificial Intelligence Technologies 2026 Summer School, *Ivan Franko National University of Lviv* — *Jun – Jul 2026*
120 hours / 4 ECTS credits. Focus: Machine learning, Python for data analysis, deep learning architectures, computer vision, cloud computing, and AI applications.

Servercamp, *Astha IT* — *May 2026 – Ongoing*
Focus: Backend development (4-month program) with C#, OOP, ASP.NET, and SQL.

Full Stack Web Development with Python, Django & React, *Ostad* — *Oct 2025 – Completed*
Focus: JavaScript, Python, HTML, CSS, Django, and React.

Industrial Robot Operation and Troubleshooting, *BITAC, Dhaka* — *Jul 2025 – Oct 2025*
Focus: Robot simulation, robot programming, troubleshooting, and Malfabasic language. *Tejgaon, Dhaka*

Competitive Programming & Extracurriculars

</> Codeforces · </> LeetCode

- Participated in Intra AUST Programming Contest, Fall 2021.
- Campus Ambassador, GUVI Geek Networks (IITM Research Park), from 2021 to 2023; organized a seminar on the importance of Data Science.